

1. What is ModelMapper and why do we use it?

Answer: ModelMapper is used to automatically convert Entity objects to DTO objects. It reduces repetitive mapping code and improves maintainability.

2. Why do we need to define ModelMapper as a Spring Bean?

Answer: Defining ModelMapper as a Spring Bean allows Spring to manage its lifecycle and inject it through dependency injection.

3. Why is constructor injection preferred?

Answer: Constructor injection ensures required dependencies are provided at object creation time and improves testability and code safety.

4. Why do we need Global Exception Handling?

Answer: Global exception handling centralizes error processing logic and provides consistent API error responses without repetitive try-catch blocks.

5. What is MethodArgumentNotValidException?

Answer: It is thrown when request body validation fails and is used to return structured field-level validation error messages.

6. Why do we use OneToMany mapping between Post and Comment?

Answer: One Post can contain multiple Comments. OneToMany mapping ensures relational consistency, supports cascading operations, and enables orphan removal.

7. Why is H2 database used during development?

Answer: H2 is an in-memory database that starts quickly, requires no installation, and is suitable for development and testing environments.

Execution Verification

```
{ "links": { "self": { "href": "http://localhost:8080/actuator", "templated": false }, "health": { "href": "http://localhost:8080/actuator/health", "templated": false, "health-path":  
{ "href": "http://localhost:8080/actuator/health/{xpath}", "templated": true } }
```

[INFO] Results:

[INFO]

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO]

[INFO]

[INFO] BUILD SUCCESS

[INFO]

[INFO] Total time: 8.477 s

[INFO] Finished at: 2026-01-26T10:15:56-08:00

[INFO]

- jdbc:h2:mem:redbook
- COMMENTS
- POSTS
- INFORMATION_SCHEMA
- Users
- H2 2.1.214 (2022-06-13)

RunRun SelectedAuto completeClearSQL statement:

Important Commands

	Displays this Help Page
	Shows the Command History
 Ctrl+Enter	Executes the current SQL statement
 Shift+Enter	Executes the SQL statement defined by the text selection
Ctrl+Space	Auto complete
	Disconnects from the database

Sample SQL Script

Delete the table if it exists	DROP TABLE IF EXISTS TEST;
Create a new table with ID and NAME columns	CREATE TABLE TEST(ID INT PRIMARY KEY, NAME VARCHAR(255));
Add a new row	INSERT INTO TEST VALUES(1, 'Hello');
Add another row	INSERT INTO TEST VALUES(2, 'World');
Query the table	SELECT * FROM TEST ORDER BY ID;
Change data in a row	UPDATE TEST SET NAME='Hi' WHERE ID=1;
Remove a row	DELETE FROM TEST WHERE ID=2;
Help	HELP ...

Adding Database Drivers

Additional database drivers can be registered by adding the Jar file location of the driver to the environment variables H2DRIVERS or CLASSPATH. Example (Windows): to add the database driver library C:/Programs/hsqldb/lib/hsqldb.jar, set the environment variable H2DRIVERS to C:/Programs/hsqldb/lib/hsqldb.jar.